AYSHA HIBA

Data Science & AI | Machine Learning, Analytics & Applied AI

UAE Residence Visa Holder | Open to Full-Time Opportunities

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PROFILE

Aspiring AI Solutions Engineer and final year M.Tech Data Science student with 2 years of software engineering experience. Skilled in Python, machine learning, data analytics, SQL, and AI/ML technologies, seeking opportunities in Data Science and AI to build impactful, data-driven solutions.

WORK EXPERIENCE

Associate Software Developer (Backend & AI) | *Cred.ai · Bengaluru* *Aug 2023 – Aug 2025*

- ***Data Quality & Validation***: Monitored and assessed data integrity across systems, implementing validation checks and debugging tools in STS to ensure consistency and reliability for analytics and AI applications.
- ***AI & Backend Development***: Developed highly scalable and performant RESTful micro-services in Java, supporting data-driven features and API endpoints.
- ***Reporting & Documentation***: Researched and documented Proof-of-Concepts (PoCs), model performance insights, and suggested improvements for code maintenance and knowledge retention.
- ***Data Governance & Standards***: Improved code quality and ensured test coverage of at least 95% in JUnit tests, upholding data integrity standards and supporting governance initiatives.
- ***Cross-Functional Collaboration***: Partnered with team members across departments, tracked and managed development tasks in Jira, and delivered solutions within specified deadlines.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C

Generative AI & LLMs: Gemini API, Prompt Engineering, RAG, Hugging Face Transformers, Text Embeddings, AI Chatbots

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, XGBoost, Random Forest, Regression, Classification, Clustering, Anomaly Detection, Time-Series Forecasting

Data Engineering & APIs: FastAPI, REST APIs, Data Preprocessing, Feature Engineering, ETL, Batch Processing

Databases & Vector DBs: MySQL, PostgreSQL, MongoDB, DynamoDB (AWS), SQL (Joins, Aggregations, Window Functions, JPQL), Chroma

Cloud & AI Platforms: AWS Cloud Essentials, Google Earth Engine, Azure Basics

Analytics & Visualization: Power BI, Streamlit, Matplotlib, Plotly, Seaborn, Advanced Excel, Dashboard Development

Development Tools: Git, Jupyter Notebook, VS Code, Postman, Google Colab, Anaconda

EDUCATION

M.Tech in CSE (Data Science) · Manipal Academy of Higher Education, Bengaluru *2025 – 2027*
CGPA 9.09

B.Tech in Computer Science & Engineering · APJ Abdul Kalam Technological University *CGPA 7.83*

CERTIFICATIONS

IBM SkillsBuild AI Practitioner
AI Ethics, ML & DL, LLM, NLP

Data Visualization with Power BI
Microsoft (Great Learning)

LANGUAGES

English – Proficient

Arabic – Basic

Hindi – Fluent

PROJECTS

Healthcare Equipment Health Analytics — *Equipment Lifecycle & Maintenance Intelligence System*
Python, FastAPI, PostgreSQL, SQLModel, Power BI

- Built an end-to-end equipment analytics platform to monitor equipment health, failures, maintenance effectiveness, replacement priorities, and service impact across healthcare facilities.
- Designed a normalized PostgreSQL database and ETL pipelines for data ingestion, validation, transformation, and management of facility and equipment data.
- Developed SQL-based KPIs and analytical views covering failure rates, lifespan, downtime, preventive maintenance, repair success, spare-parts availability, and replacement priority.
- Built an interactive Power BI dashboard for equipment health monitoring, maintenance insights, replacement planning, and service impact analysis; developing a natural-language RAG and text-to-SQL layer for querying equipment data.

Enterprise AI Knowledge Assistant — *Dubai Property Law RAG System*
Python, FastAPI, ChromaDB, Sentence-Transformers, Groq, Streamlit

- Built an end-to-end retrieval-augmented generation system that answers questions from Dubai real estate legislation with article-level citations.
- Ingested and indexed 5 statutory documents (450+ chunks) from official DLD/Dubai Legislation Portal sources, with PDF parsing, overlap chunking, and vector storage in ChromaDB.
- Implemented multi-provider LLM abstraction with automatic model failover and retry-on-rate-limit, keeping generation decoupled from retrieval.

Marketing Campaign Intelligence Dashboard — *BI & Decision Support*
MySQL, SQL, Power BI, Data Cleaning, Data Analysis, Data Visualization, Marketing Analytics

- Built an end-to-end analytics application combining SQL, Power BI, and Python to analyse campaign performance, channel effectiveness, customer engagement, and revenue outcomes.
- Designed advanced DAX measures and business KPIs including ROI, ROAS, CTR, and conversion metrics.
- Designed interactive Power BI dashboard with dynamic filters, KPI cards, trend analysis, channel comparisons, and campaign-level performance views, enabling stakeholders to identify high-performing campaigns and optimization opportunities.

FinOps Copilot — *AI-Powered Cloud Cost Optimization*
Python, Streamlit, ChromaDB, scikit-learn, Gemini API, AWS

- Built an AI-powered cloud cost optimization assistant using Streamlit, machine learning, and LLM APIs to help users analyze AWS spending through natural language.
- Implemented a RAG pipeline over 50K+ AWS pricing records using Chroma vector store and text embeddings, enabling grounded, context-aware responses to cost queries.
- Integrated anomaly detection, bill prediction, and conversational querying, combining ML models with Generative AI (Gemini API) to deliver explainable recommendations.

Urban Carbon Emission Forecasting (Bengaluru) — *Applied AI & Data Science Research*
Python, Google Earth Engine, XGBoost, Markov Chain Modelling, Geospatial Analysis

- Developed a Hybrid Markov Chain–XGBoost framework to classify Bengaluru wards into carbon sinks, neutral zones, and emitters and model their year-to-year transitions using multi-source satellite data.
- Built Python and Google Earth Engine data pipelines for preprocessing, feature engineering, temporal analysis, and ward-level spatial aggregation.
- Applied XGBoost and explainable AI (SHAP) to predict emission intensity and identify key environmental and anthropogenic drivers influencing model predictions.